

Erin Avllazagaj

PhD Candidate @ [Maryland Cybersecurity Center \(MC2\)](#) | albocode@umd.edu | <https://albocoder.github.io/blog/>

RESEARCH INTERESTS

My broad research interests cover data-driven malware analysis and **automatic exploitation**. I'm currently more focused in automating the exploitation steps for the Linux kernel. Specifically, in my recent PhD work I have developed a tool (called *SCAVY*) to automatically find kernel objects that when overwritten by memory corruptions bugs can reach privilege escalation. My participation in various CTF competitions and my internship work at Meta have been major influences in this new research direction as well as my interest in sharing exploitation writeups in [my blog](#).

EDUCATION

University of Maryland

Ph.D. student in Electrical and Computer Engineering, Advisor: [Prof. Tudor Dumitras](#)

College Park, MD

Aug. 2018 – Present

Bilkent University

B.S. degree in Computer Science

Ankara, Turkey

Aug. 2014 – May 2018

- Scholarly paper: “Privacy-Related Consequences of Turkish Citizen Database Leak” (**INFER’16**)
 - * Advisors: [Prof. Erman Ayday](#), [Prof. Ercüment Çiçek](#)
- Senior Project: “[Andlit](#)”
 - * Advisor: [Prof. İbrahim Körpeoğlu](#)
 - * The project is a social network of AI models that perform face recognition in real-time. It’s designed to help the visually impaired recognise their environment, the friends and the *friends of their friends*. The project was called a “third eye” by various Turkish media agencies [1*] [2] [3] [4] [5] including [daily newspapers](#).

PROFESSIONAL EXPERIENCE

[Meta \(Product Security Team\)](#) | *PhD SWE Intern* | *California, USA*

June 2022 – September 2022

- I worked with my manager Dr. David Molnar and my peers to migrate the an old Zoncolan static analysis pipeline to a new pipeline for the Privacy team to use. During this time I learned more about big data management tools like Dataswarm, Hive, Spark etc.

[EURECOM](#) | *Visiting Scientist* | *Nice, France*

July 2018 – August 2018

- I worked with Dr. Leyla Bilge and Dr. Davide Balzarotti to analyze and understand malware behavior data in the wild. I was able to list and solve the problem of noise in the data and provide meaningful case studies for each.

[Maryland Cybersecurity Center](#) | *Research Intern* | *Maryland, USA*

June 2017 – August 2017

- I worked with Dr. Octavian Suci to investigate code reuse of exploits in the wild. In this work I read many papers and used their static analysis techniques to measure closeness of code snippets parsed with Joern.

[4S \(contractor for HavelSan\)](#) | *Red Teaming* | *Ankara, Turkey*

July 2016 – August 2016

- I was tasked with testing a Unity based game for security issues. The game was a red-blue team simulation. The testing was important due to the client-server architecture and the security implications the server would impose at HavelSan. Without access to the source code I used tools from DevXUnityUnpacker, to Cheat Engine and IDA Pro.

PUBLICATIONS

Conferences

E. Avllazagaj, Y. Kwon, T. Dumitras, “SCAVY: Automated Discovery of Memory Corruption Targets in Linux Kernel for Privilege Escalation”, In *USENIX Security 24*, 2024

E. Avllazagaj, Z. Zhu, L. Bilge, D. Balzarotti, T. Dumitras, “When Malware Changed Its Mind: An Empirical Study of Variable Program Behaviors in the Real World”, In *30th USENIX Security Symposium (USENIX Security 21)*, 2021

* **1st place winner** of CSAW21’ Applied Research competition [[certificate](#)]

* Press highlights: [[CSO-DE](#)] [[Cyberwire](#)] [[CSO-EN](#)] [[UMIACS](#)]

Workshops

E. Avllazagaj, E. Ayday, E. Çiçek, “Privacy-Related Consequences of Turkish Citizen Database Leak”, In International Workshop on Inference & Privacy in a Hyperconnected World (INFER’ 16), 2016

N. Garg, I. Shahid, **E. Avllazagaj**, J. Hill, J. Han, N. Roy, “Thermware: Toward side-channel defense for tiny iot devices”, The 24th International Workshop on Mobile Computing Systems and Applications (HotMobile’ 23), 2023

Posters

O. Suci, **E. Avllazagaj**, T. Dumitras, “The Secret Life of Pwns: Characterizing and Predicting Exploit Weaponization”, In Conference on Applied Machine Learning for Information Security (CAMLIS’ 19), 2019

HONORS AND AWARDS

Fellowships

Clark Doctoral Fellowship GRA/GTA support (4 years)	2018 – 2022
Clark Doctoral Fellowship stipend (first 2 semesters)	2018 – 2019
Clark Doctoral Fellowship travel grant (4 years)	2018 – 2022
Bilkent University Full Tuition+Accommodation scholarship (all 4 years)	2014 – 2018
Mehmet Akif High School Tuition scholarship (all 3 years)	2011 – 2014

Awards

1 st place in CSAW21’ Applied Research Competition certificate	
2 nd place in Informatics Olympiad of Tirana	2013

SERVICES

Program Committee

[Full PC Member] CSAW 22’ 23’	2022, 2023
[Artifact Evaluation] Usenix Security	2021 – 2024
[Artifact Evaluation] PoPETS (Accepted papers from PETS)	2021 – 2023
[Journal Reviewer] Computer Science Review Journal	2021
[Journal Reviewer] IEEE ISCC	2022

External reviewer

Network and Distributed System Security Symposium (NDSS)	2020
USENIX Security Symposium (USENIX)	2019,2023
International Symposium on Research in Attacks, Intrusions and Defenses (RAID)	2018, 2019
ACM Conference on Computer and Communications Security (CCS)	2019, 2020
IEEE Security and Privacy (S&P)	2022, 2023